CCNA 200-301 V2.0

# 05 IPv6 Addressing

Part I — Foundations

EXAM TOPICS

1.4


LECTURER

**Dr. Mustafa Hameed**

اسلامیہ یونیورسٹی بہاولپور

DEPARTMENT OF INFORMATION TECHNOLOGY

THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

# Where this chapter sits

| | | |
|---|---|---|
| **Part I** | **Foundations** | chapters 1–7 |
| **Part II** | Switching and Network Access | chapters 8–14 |
| **Part III** | IP Routing | chapters 15–19 |
| **Part IV** | Network Services and Security | chapters 20–26 |
| **Part V** | Wireless, Virtualisation and Operations | chapters 27–32 |
| **Part VI** | Sitting the Exam | chapters 33–34 |

# What you will be able to do

- **Abbreviate** and expand an IPv6 address correctly, applying both compression rules without ambiguity.
- **Identify** the address types the exam names: global unicast, link local, unique local, multicast, and the solicited-node address.
- **Construct** an interface identifier by modified EUI-64, which the blueprint calls out explicitly.
- **Size** a prefix, and **diagnose** the assignment faults that stop an IPv6 host communicating.

ASSUMED KNOWLEDGE

# Before we start

Chapter 4 — the ideas of network part, host part and prefix length carry over unchanged. Chapter 3 for MAC addresses, which EUI-64 is built from.

KEY TERMS

# Vocabulary for this chapter

- global unicast
- link local
- unique local
- multicast
- solicited-node
- anycast
- modified EUI-64
- SLAAC
- router advertisement
- neighbour discovery
- dual stack
- prefix

Each is defined where it first matters — not here.

ON THE EXAM

# What v2.0 asks, and what it dropped

**This is what the blueprint actually asks for**

Topic **1.4** says *troubleshoot* IPv6 address configuration, assignment and prefix sizing, and narrows the scope to “unicast and modified EUI 64”. The previous blueprint asked you to compare every address type in detail; this one does not. Concentrate on getting an address onto an interface, understanding where the link-local one came from, and spotting a wrong prefix.

# Writing the address

CONCEPT

# The two compression rules

- **Drop leading zeros** in any group. `0db8` becomes `db8`; `0000` becomes `0`. Trailing zeros are never dropped.
- **Replace one run of all-zero groups with `::`**. You may do this **once only**, because a second `::` would make the address ambiguous — there would be no way to know how many zeros each stood for.

WORKED EXAMPLE

# Compressing an address

Start with the full form:

`2001:0db8:0000:0000:0000:ff00:0042:8329`

- Drop leading zeros in each group: `2001:db8:0:0:0:ff00:42:8329`
- Replace the longest run of zero groups with `::`: `2001:db8::ff00:42:8329`

Expanding is the same in reverse. Given `2001:db8::1`, count the groups present — two before the `::` and one after, so three — and the `::` must therefore stand for five zero groups: `2001:0db8:0000:0000:0000:0000:0000:0001`.

WATCH OUT

# There is no broadcast in IPv6

IPv6 has no broadcast address at all. Where IPv4 would broadcast, IPv6 uses a multicast group — usually `ff02::1`, all nodes. This is why ARP does not exist in IPv6: its job is done by neighbour discovery using the solicited-node multicast address, which reaches only the handful of hosts whose addresses end the same way rather than every device on the segment.

# Modified EUI-64

CONCEPT

# Turning a 48-bit MAC into a 64-bit interface ID

- Split the MAC in half.
- Insert `fffe` in the middle.
- **Flip the seventh bit** of the first byte — the universal/local bit. In hex this means adding or subtracting 2 from the second hex digit.

WORKED EXAMPLE

# EUI-64 from 00d0.58a1.0201

- Split: `00d0.58` and `a1.0201`.
- Insert `fffe`: `00d0:58ff:fea1:0201`.
- Flip the seventh bit of `00`. In binary `00` is `0000 0000`; flipping the seventh bit gives `0000 0010`, which is `02`.
- Interface ID is `02d0:58ff:fea1:0201`. With a prefix of `2001:db8:acad:10::/64` the full address is `2001:db8:acad:10:2d0:58ff:fea1:201`.

The quick check: an address containing `ff:fe` in the middle of its host portion was almost certainly built by EUI-64.

# Configuring IPv6

CONFIGURE

# IPv6 on a router interface, three ways

IOS · configuration mode

```
! Routing must be enabled globally first -- this is the step most often
! forgotten, and without it the router will not forward or advertise.
ipv6 unicast-routing
!
interface GigabitEthernet0/0
 description User VLAN 10 gateway
 ! 1. A fully specified address.
 ipv6 address 2001:db8:acad:10::1/64
 ! 2. Or let the device build the host part by EUI-64.
 ipv6 address 2001:db8:acad:20::/64 eui-64
 ! 3. A link-local address is created automatically, but you can pin it
 !    to something memorable -- worth doing, because routing protocols
 !    show it as the next hop.
 ipv6 address fe80::1 link-local
 no shutdown
```

VERIFY

# show ipv6 interface brief

```
R1#

GigabitEthernet0/0     [up/up]
    FE80::1
    2001:DB8:ACAD:10::1
    2001:DB8:ACAD:20:2D0:58FF:FEA1:201
GigabitEthernet0/1     [administratively down/down]
    unassigned
```

CONCEPT

# How a host gets an address: SLAAC versus DHCPv6

A host sends a router solicitation to `ff02::2`; a router replies with a **router advertisement** carrying the prefix. Then either:

- **SLAAC** — the host builds its own address from the advertised prefix plus an interface ID it generates. No server involved.
- **DHCPv6** — the advertisement tells the host to ask a server, which supplies the address and other options.

A host with only a `fe80::` address and no global one has heard no router advertisement. That is the IPv6 equivalent of an APIPA address, and it points at the router, not the host.

# Common pitfalls

- **Omitting `ipv6 unicast-routing`.** Interfaces come up and addresses look right, but the router neither forwards nor advertises.
- **Using `::` twice.** It is not merely discouraged, it is invalid, because the address can no longer be expanded uniquely.
- **Dropping trailing zeros.** `2001:db8:1000::` cannot be shortened to `2001:db8:1::`. Only *leading* zeros go.
- **Assuming a prefix other than /64 for a LAN.** SLAAC requires a 64-bit interface ID, so a LAN prefix longer than /64 breaks autoconfiguration. Point-to-point links are the usual exception.

# Dual-stack host reaches IPv4 destinations only.

A workstation has a working IPv4 address and, for IPv6, only `fe80::2d0:58ff:fea1:201`. It cannot reach any IPv6 destination, on or off the segment.

# Dual-stack host reaches IPv4 destinations only.

**What would you check first — and what do you expect to see?**

# Why it is doing that

**Diagnosis.** A link-local address is generated by the host itself and proves nothing except that the interface is up. The absence of a global address means no router advertisement is being received. Either no router is configured for IPv6 on that VLAN, or `ipv6 unicast-routing` is missing, or the router interface has no global prefix to advertise.

# And the lesson

**Fix.** On the gateway, confirm `ipv6 unicast-routing` is present in the running configuration, then check `show ipv6 interface GigabitEthernet0/0` shows a global address and that router advertisements are not suppressed. Bring the prefix back and the host will build its own address within seconds — no change is needed on the host at all.

# Dual-stack a small network

**Topology.** One router, one switch, two PCs, addressed for both protocols.

1. Enable `ipv6 unicast-routing` and give `Gi0/0` the address `2001:db8:acad:10::1/64` and link-local `fe80::1`.
2. Set both PCs to obtain IPv6 automatically. Record the addresses they build and confirm they contain `ff:fe`.
3. Compute by hand what each PC's EUI-64 address should be from its MAC, and check your answer against the device.
4. Ping between the PCs using IPv6, then ping the router's link-local address from a PC. Note that the link-local ping needs no global addressing at all.

# Dual-stack a small network (continued)

5. Remove `ipv6 unicast-routing` and observe what happens to the PCs after their addresses expire.
6. **Extension.** Configure a second VLAN with prefix `2001:db8:acad:20::/64` and route between the two.

CHECKPOINT 1 of 3

# Compress `2001:0db8:0000:0012:0000:0000:0000:00ab` fully.

Discuss · then advance

ANSWER

# Checkpoint 1 of 3

**Compress `2001:0db8:0000:0012:0000:0000:0000:00ab` fully.**

> `2001:db8:0:12::ab`. Leading zeros drop in every group; the single run of three all-zero groups becomes `::`. Note the fourth group is `0`, not part of the `::` — only one run may be compressed, and the longer one wins.

CHECKPOINT 2 of 3

# A host shows only an `fe80::` address. What is missing, and on which device would you look?

Discuss · then advance

ANSWER

# Checkpoint 2 of 3

## A host shows only an `fe80::` address. What is missing, and on which device would you look?

> A global unicast address, which means no router advertisement is being received. Look at the router or layer 3 switch for the segment: is `ipv6 unicast-routing` enabled, and does the interface have a global address to advertise a prefix from?

CHECKPOINT 3 of 3

# Build the modified EUI-64 interface identifier for MAC `aabb.cc00.0100`.

Discuss · then advance

ANSWER

# Checkpoint 3 of 3

## Build the modified EUI-64 interface identifier for MAC `aabb.cc00.0100`.

> Split the MAC, insert `fffe`, and flip the seventh bit: `aa` becomes `a8`. The result is `a8bb:ccff:fe00:0100`.

# What to take away

- Drop leading zeros; use `::` once, for the longest zero run.
- `2000::/3` global, `fe80::/10` link local, `fd00::/8` unique local, `ff00::/8` multicast. There is no broadcast.
- EUI-64: split the MAC, insert `fffe`, flip the seventh bit.
- `ipv6 unicast-routing` is a separate, easily forgotten step.
- LANs are /64 because SLAAC needs a 64-bit interface ID.
- Only a link-local address means no router advertisement was heard — investigate the router, not the host.

# Where we got to

- Drop leading zeros; use `::` once, for the longest zero run.
- `2000::/3` global, `fe80::/10` link local, `fd00::/8` unique local, `ff00::/8` multicast. There is no broadcast.
- EUI-64: split the MAC, insert `fffe`, flip the seventh bit.
- `ipv6 unicast-routing` is a separate, easily forgotten step.

NEXT

**Chapter 6 — TCP, UDP and the Diagnostic Toolkit**